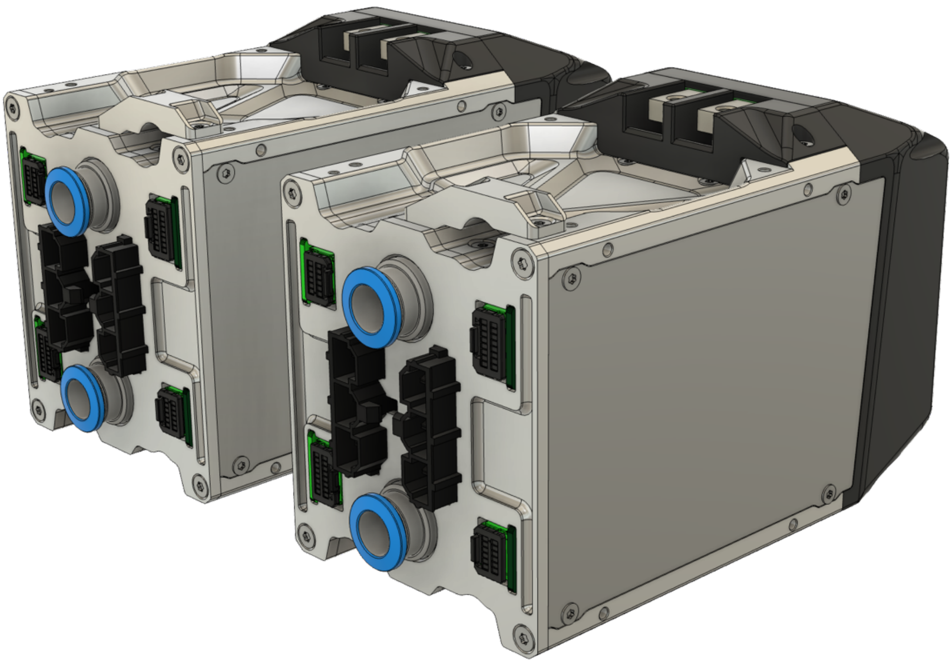
DTI FORMULA STUDENT INVERTER

Application manual for F-SIC inverter

Version 1.8

Valid for F-SIC-A inverter



Please read the user manual carefully before using the controller. Electrical systems can cause danger to humans, property and nature; therefore, precautions shall be taken to avoid any risk.

Table of contents

1. HISTORY AND RELATED DOCUMENTS

1.1 Document history

8/2024	**V1.0** Document created; basic specifications added mainly with simulations.
10/2024	**V1.2** Extended with measurements. AC connector change. HEIDENHAIN encoder hardware extension
01/2025	**V1.3** Data correction for release, related documents added
06/2025	**V1.4** Pinout description change
07/2025	**V1.5** Low Voltage Harness Pinout fix, Pin2 to GND from 12_VIN
08/2025	**V1.6** Sensor connector pinout fix, from pin 16 to 20
09/2025	**V1.7** Cooling connector recommendation change

1.2 Related documents

- DTI Can manual
- DTI-COM manual
- F-SIC-A Inverter Specification
- F-SIC-A Inverter Drawing
- F-SIC-A Inverter CAD Model
- F-MOT Position sensor connector assembly Drawing
- F-SIC-A Inverter Scope of delivery

2. LIABILITY AND SAFE USE OF THIS UNIT

DTI hardware, DTI CAN Tool and the DTI firmware are experimental products designed to develop, evaluate and test electrical systems incorporating electric motors or actuators. Electrical systems can cause danger to humans, property and nature; therefore, precautions shall be taken to avoid any risk. Under no circumstances shall the device be used where humans or property are put to risk without thoroughly validating and testing the whole system. Software and hardware interact in various ways, and developers cannot foresee all possible combinations of hardware used together with software, nor problems that can occur in these different combinations. Can Tool and the DTI firmware are experimental software designed to develop and test. Electrical systems can cause danger to humans, property and nature; therefore, precautions shall be taken to avoid any risk.

Only trained and educated personnel are authorized to configure, integrate and operate the device who know the potential risks, and can act in case of any misbehavior, or fault that occours.

The integrator has the responsibility to select the proper hardware components for the specific use-case. DTI takes no responsibility for any issues, problems, failed examinations, failed registration procedures or evaluations due to inappropriate configuration, system design or general system structure or lacking documents, certifications for the specific area-of use.

Things that can happen, even when using the correct settings, are

- electrical failure
- fire
- electric shock
- hazardous smoke
- overheating motors and actuators
- overloaded power sources, causing fire or explosions (e.g., Lithium-Ion Batteries)
- motors or actuators stopping from spinning/moving
- motors or actuators locking in, acting like a brake (full stop)
- motors or actuators losing control over torque production (uncontrolled acceleration or braking)
- interferences with other systems
- other non-intended or unforeseeable behaviour of the system

DTI CAN Tool and the DTI firmware are developer software that for safety reasons may only be used

- by experts and experienced users, knowing exactly what they do.
- following safety standards applicable in the area of usage.
- under safe conditions where software or hardware malfunction will not lead to death, injuries or severe property damage.
- keeping in mind that software and hardware failures can happen. We can't provide any kinds of warranty because every system is unique and we cannot make sure its safety and integrity. Although we design our products to minimize such issues, you should always operate with the understanding that a failure can occur any time without warning. As such, you shall take the appropriate precautions to minimize danger in case of failure.

Each and every inverter is being tested as a part of the manufacturing progress. DTI assumes no liability in case a customer uses system components for the purposes that they have not been developed for or tested.

DTI reserves the right to change any information within the scope of this whole manual. All connection circuitry described is meant for reference only, are theoretical circuits and not mandatory. DTI does not assume any liability, expressively or inherently, for the information contained in this document, for the functioning of the device or its suitability for any specific application.

2.1 Discharge

Each Controllers are equipped with internal discharge resistors for improved safety. The discharge time for the inverter's internal capacitors is **3 minutes**. The discharge resistance is **330 KOhm.**

> The inverter's internal discharge circuit is a safety feature, and it is not meant to replace the system discharge circuitry.

> The minimum wait time before disconnecting the HV cables from the inverters is 5 minutes after powering off the High voltage. In case the safety time frame is not kept correctly, the charged DC capacitor's voltage can be a source of a potential electric shock which can be dangerous or even lethal.

3. WARNING SIGN EXPLAINED

The Warning sign contains the most important safety notices that has to be taken into consideration by everyone who approaches the inverter. The sign cannot be removed or covered off he controller.

In case of the sign is being damaged, or detached from the inverter, please get in touch with DTI as soon as possible at the availability mentioned on the cover page.

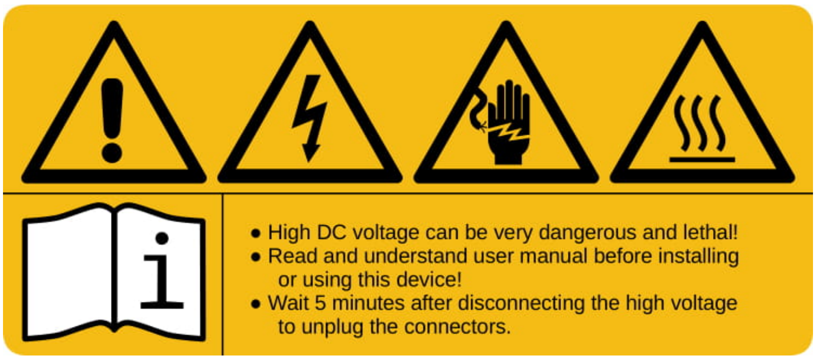
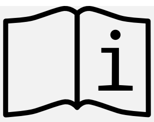


Figure 1. Safety note

Table 1. Safety notes explained

Description	
	Please read this manual carefully, with particular regard to safety notes.
	**Danger of electric shock** Touching HV wires and HV connectors are prohibited without ensuring that the HV is disconnected.
	**Danger of high voltage** The device is attached to high voltage.
	**Danger of hot surfaces** The inverter case, and /or the coolant fluid can get hot during and after operation, touching any of these can cause burn damage.
	**General Warning** Only trained and educated personnel can do integration and maintenance.

4. OVERVIEW

Key Features

- 36-620V power stage input range
- 8-20V logic stage input range
- 40/80 Arms current rating
- Up to 200.000 eRPM motor speed
- Multiple controlling peripherals
- CAN (ISO 11898-2)
- Multiple controlling methods
- Current (Torque) mode
- Speed control mode (eRPM)
- FOC motor control
- 8-30kHz Switching frequency using central aligned PWM
- Regenerative braking, generator mode
- Motor angle positioning
- SSI, ABI, Resolver, EnDat motor position support
- Hardware and software overvoltage, overcurrent protections
- Undervoltage protection and limitation
- Power stage and Motor temperature protection and limitation
- Active Junction temperature measurement and limitation
- Motor Speed limitation
- Motor Power limitation
- Position sensor error sensing
- AC and DC current limitation
- Configurable digital Input (i.e., Inverter enable, reverse switch, etc, custom drive mode triggering)
- Configurable digital Output (i.e., RPM output for VDO instruments, Error output, limiting feedback)
- Auto-Tune for motor position sensors
- Automatic motor identification
- Magnetic field weakening and MTPA

Application

- Formula Student 2/4WD drives
- Prototype small cars, motorcycles, karts
- Battery powered machines

Description

The F-SIC is an inverter specifically developed for Formula Student teams, considering the current rules of the competition. It houses two completely independent inverters in one casing. It features integrated water cooling, which is also shared per unit. The design goal was to achieve the best possible weight/power ratio, which was made possible by using SiC semiconductors, high switching frequency, and 3D printed cooling plate. The inverter is designed to be placed inside a compartment, such as the PDU box, for better integration. For this reason, the design is not waterproof and does not include interlock connectors.

4.1 Nameplate explained

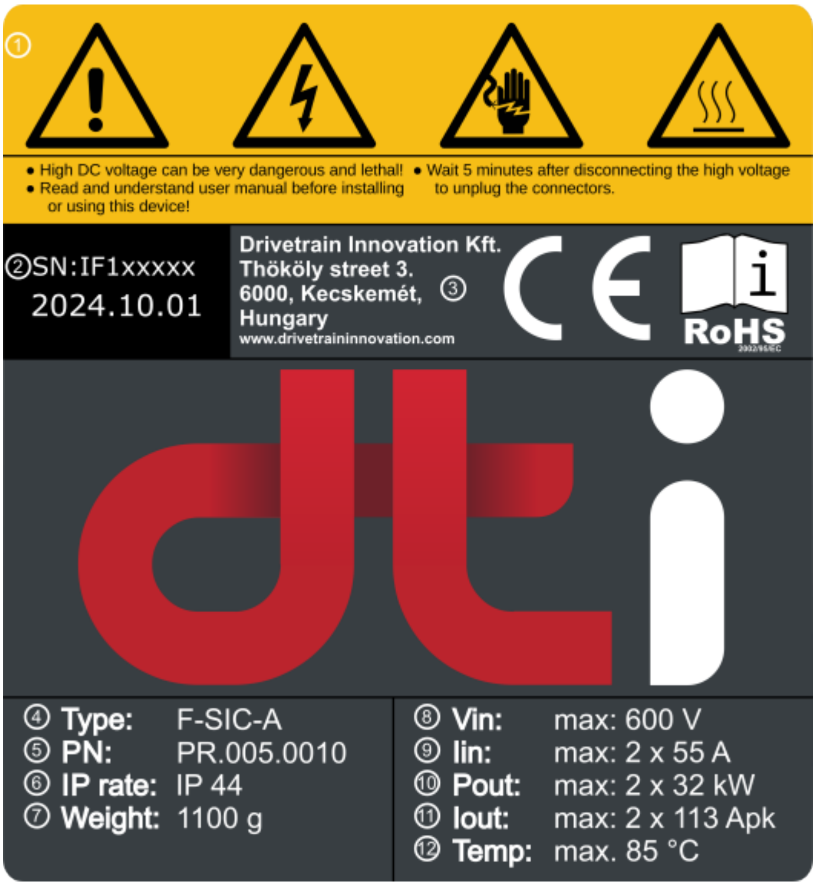


Figure 2. Nameplate

Table 2. F-SIC-A Nameplate explained

No.	Description	No.	Description
1.	Potential hazards	8.	Input voltage rating
2.	Serial number	9.	Input DC current rating
3.	Manufacturer information	10.	Maximum power output
4.	Inverter type	11.	Maximum peak current
5.	Part number	12.	Maximum working temperature
6.	IP protection rating		
7.	Product net weight		

5. MECHANICAL DRAWINGS

Dimensions are shown in mm.

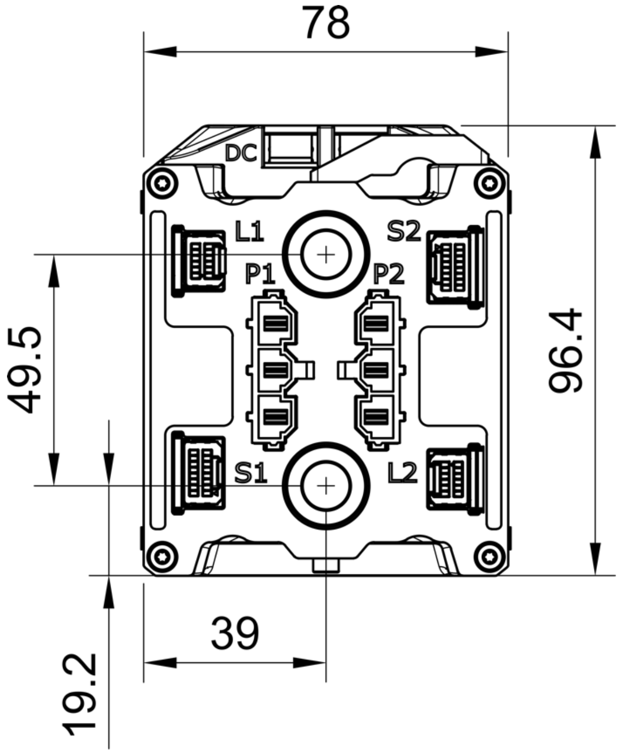


Figure 3. Front side mechanical drawing

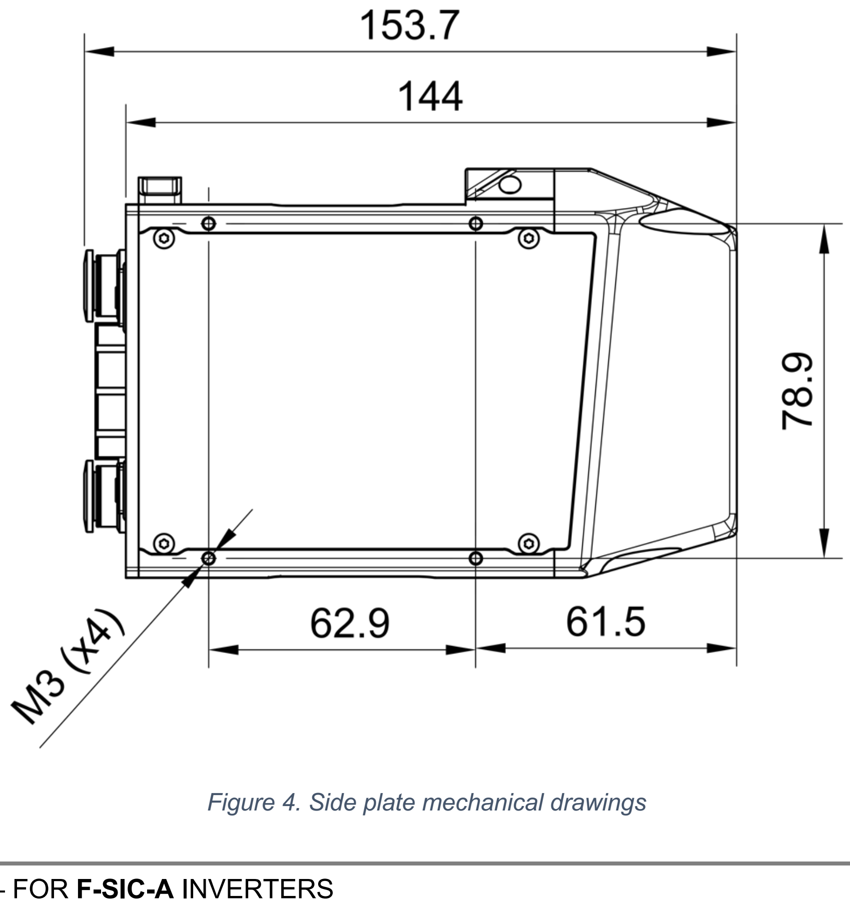


Figure 4. Side plate mechanical drawings

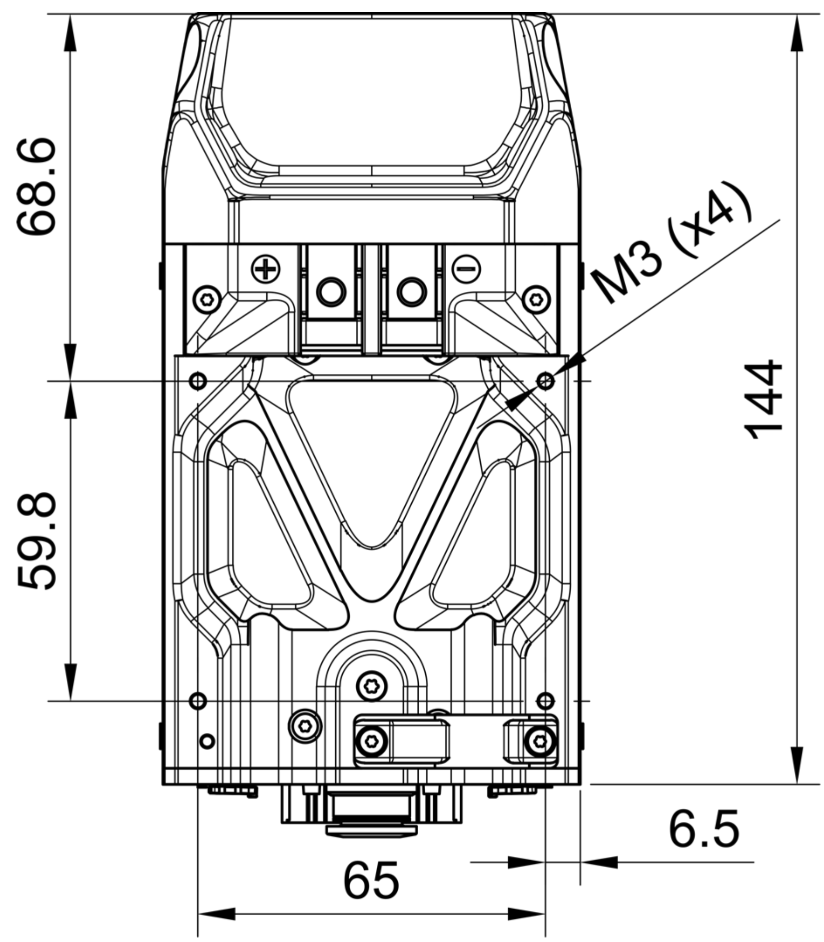


Figure 5. Top side mechanical drawings

6. COMPATIBLE MOTOR POSITION SENSORS

6.1 Encoders

Supported Encoders: **RLS RM44SI** (SSI + Incremental) 10B,11B,13B resolution variants. Other ABI+ SSI type position sensor variants may work; however, those have not been tested. When choosing this encoder type, you must make sure that both ABI and SSI interfaces are available, as the inverter uses both for fully functioning operation.

- Compatible with normal or differential signals.
- ±15 kV ESD-Protected signals
- Maximum input signal voltage: 5 V
- Maximum encoder resolution: 8192 (counts per revolution)
- Maximum input frequency: 400 kHz

> The RLS RM44 encoders are metal-housed encoders equipped with shielded cables. It is important to connect the metal housing of the encoder and the shielding of the cable to ground at only one point. This point is recommended to be near the inverter. In this case, the encoder must be electrically isolated from the motor.

6.2 Resolver

The controller is compatible with various type resolvers. The tested, recommended variants:

- TAMAGAWA Smartsyn **TS2620N21E11.**
- TE Connectivity **2-1393047-5**

> Tamagawa resolvers must be equipped with shielding cap. It is important to use shielded cables.

6.3 Other digital sensors

The inverter has a digital data and clock peripheral, which makes it compatible with other types of devices without hardware modification. **These will be implemented in the future and will become available through a software update. For the actual stage of the development get in touch with DTI support via support@drivetraininnovation.com.**

- RS485
- BiSS C
- EnDat 2.1 / 2.2

> The use of differential signals is mandatory in all cases
>
> DTI reserves the right not to release the feature across all hardware revision variants.

7. ELECTRICAL CONNECTIONS

7.1 Connector identification

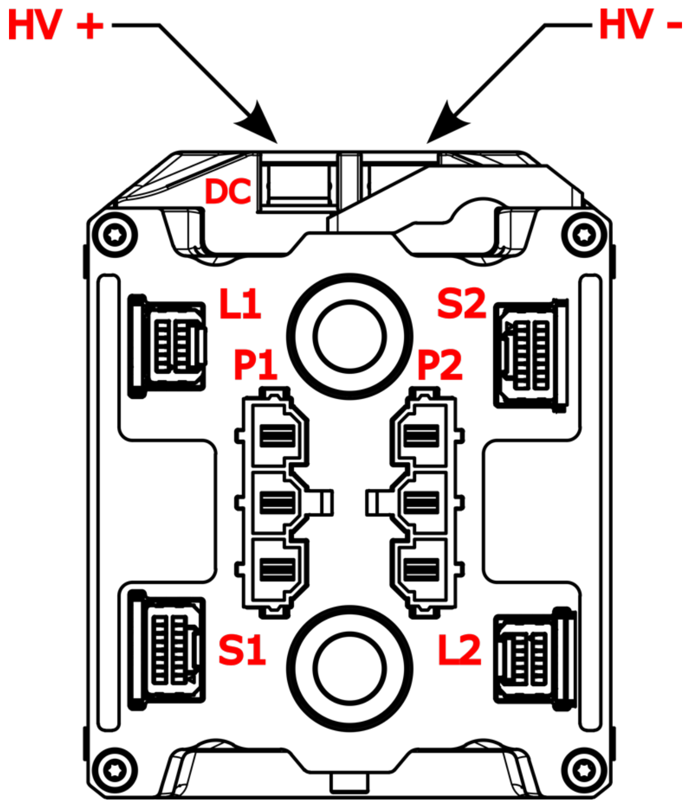


Figure 6. Electric connector identification

Table 3. Electrical connection

Pin name	Pin description
DC	HV DC connector
L1	**Inverter 1.** LV harness connector
S1	**Inverter 1.** motor position connector
P1	**Inverter 1.** HV AC connector
L2	**Inverter 2.** LV harness connector
S2	**Inverter 2.** motor position connector
P2	**Inverter 2.** HV AC connector

7.2 Harness connector pinout (L)

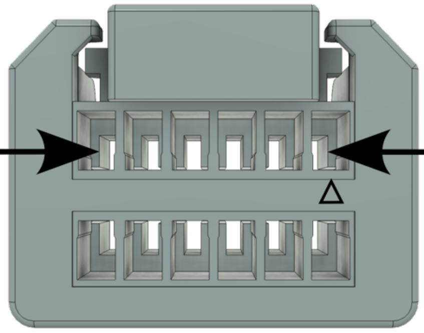


Figure 7. Low voltage harness connector

Connector type:	MOLEX **5054321201**
Terminal LOCK	MOLEX **2145282060**
28 AWG crimp type:	MOLEX **505431-1100**
Recommended wire:	ÖLFLEX® **HEAT 260 SC 28 AWG**

Table 4. Harness connector pinout description

Id	Pin name	Pin description
1	12_VIN	Auxiliary voltage plus
2	GND	Auxiliary voltage plus
3	D_INPUT	Digital input, internal pullup resistor
4	D_OUTPUT	Open Drain output, maximum 240 mA
5	CAN2_L	CAN Low
6	CAN1_L	CAN Low
7	CAN2_ H	CAN High
8	CAN1_H	CAN High
9	GND	Auxiliary voltage minus
10	GND	Auxiliary voltage minus
11	12_VIN	Auxiliary voltage plus
12	GND	Auxiliary voltage minus

7.3 Motor sensor connector pinout (S)

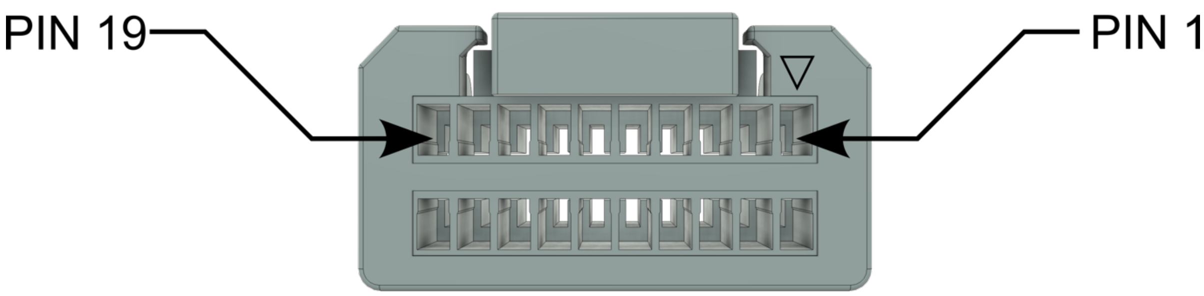


Figure 8. Motor position sensor connector

Connector type:	MOLEX **5054322001**
Terminal LOCK	MOLEX **2145282100**
28 AWG crimp type:	MOLEX **505431-1100**
Recommended wire:	ÖLFLEX® **HEAT 260 SC 28 AWG**
Recommended cable:	Amphenol **MSXS08ML-SXSML-SXXXX**

Table 5. Position sensor connector pinout description

Id	Pin name	F-Sic (FMOT-A)	Pin description
1	TEMP Pin 1	WHT/BROWN	Motor temperature sensor 1
2	TEMP Pin 2	BROWN	Motor temperature sensor 2
3	ENC_A+		Incremental encoder, differential A+ signal input
4	ENC_B-		Incremental encoder, differential B- signal input
5	ENC_A-		Incremental encoder, differential A- signal input
6	ENC_B+		Incremental encoder, differential B+ signal input
7	ENC_NM+		Incremental encoder, differential NM+ signal input
8	ENC_NM-		Incremental encoder, differential NM- signal input
9	RES_COS-	ORANGE	Resolver, differential COS- signals input
10	RES_SIN+	WHT/GREEN	Resolver, differential SIN+ signals input
11	RES_COS+	WHT/ORANGE	Resolver, differential COS+ signals input
12	RES_SIN-	GREEN	Resolver, differential SIN- signals input
13	RES_EXC+	WHT/BLUE	Resolver, differential EXC+ signals output
14	ENC_DATA+		Digital encoder, differential DATA+ signal
15	RES_EXC-	BLUE	Resolver, differential EXC- signals output
16	ENC_DATA-		Digital encoder, differential DATA- signal
17	5_VOUT		Sensor supply V+ (+5VDC, maximum 200mA)
18	ENC_CLK+		Digital encoder, differential CLK+ signal
19	GND		Sensor supply V-
20	ENC_CLK-		Digital encoder, differential CLK- signal

7.4 DC High voltage connection

The thread size on the connection surface is M5. A cable lug must be used that can handle the combined DC current of both inverters. Additionally, the wire and the cable lug must withstand a higher current load than the DC fuse assigned to this line.

Cable lug: Würth WA-CLUG **5580506**

Recommended cable: Coroflex **9-2641 (2 x 6,0 mm²)**

> Always use insulating tubing on the cable lugs. The maximum tightening torque of the screw is 2Nm.

7.5 AC High voltage connection

The connector is compatible with (equivalent to 1-8 mm²) wires. Due to the typical load profile, we recommend using a 2.5 mm² cross-sectional wire.

Connector type: Molex **428160312**

Crimp terminal: Molex **428150042**

Recommended cable: Coroflex **9-2641 (3 x 2,5 mm²)**

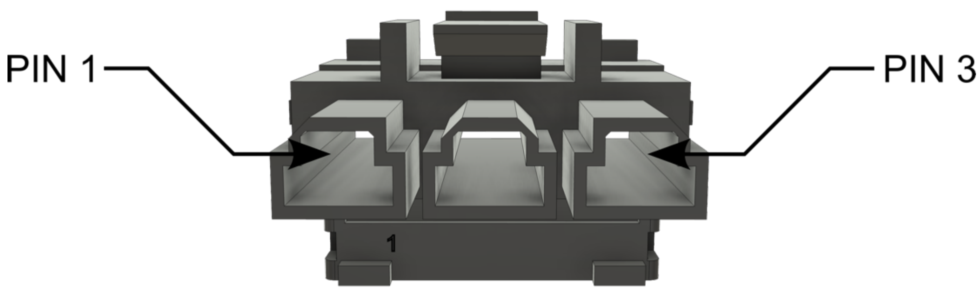


Figure 9. Motor phase AC connector

Table 6. AC high voltage connector pinout description

Id	Pin name	Pin description
1	W	Motor high voltage W phase
2	V	Motor high voltage V phase
3	U	Motor high voltage U phase

7.6 Grounding

> The controller casing must be electrically connected to the chassis **which is the potential of the low voltage ground**.

The outer surface of the inverter has all metal points suitable for grounding the shielding. A total of 16 pieces M3 threaded holes are located on the inverter's housing, which can be used for mounting and shielding purposes. Fix the inverter with at least 4 points onto a metal surface. If you are using two inverter blocks, mount both inverters on a common metal surface. This metal surface can serve as the common star point for both high and low voltage shielding and the LV GND.

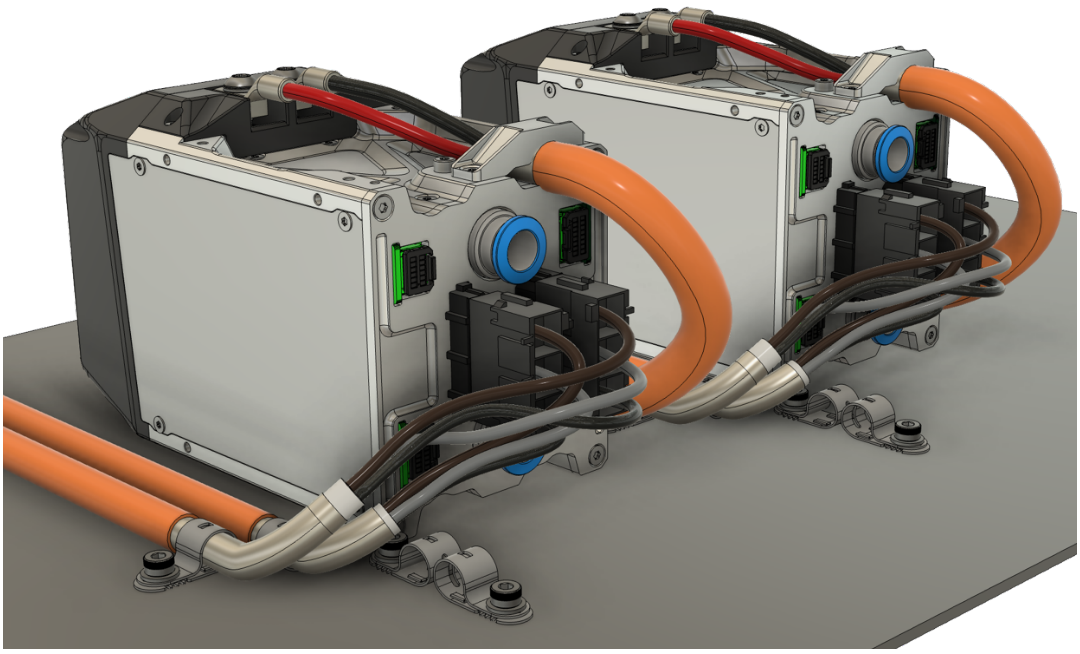


Figure 10. High voltage cable grounding example

The high-voltage cables should be grounded at a single point on the inverter side. A good practice is to use the **WE-EEL Earthing Metal Clip** components from Würth. This allows for the creation of a strong grounding point.

8. PC CONNECTION AND CONTROL

To carry out the system integration, the inverter needs to be set-up and configured according to the system's specifications. Also, a position sensor auto-tune needs to be initiated to calibrate the motor position sensor with the electric motor.

The above tasks can be done via a PC running at least Windows 11 *(for the best user experience at least an Intel i3 or AMD equivalent CPU is recommended)* with an USB 2.0 connector and a DTI-COM Diagnostic tool.

Use DTI CAN Tool application to interface the Controller, offering different features:

8.1 DTI CAN Tool

The CAN Tool has various additional features for an easier set-up and tuning. Such features are:

- Real time data analysis with plots of various parameters of the inverter
- Visual feedback of the most basic parameters
- Fault code extraction
- Load and save configurations
- Operation by keypad for system evaluation with precautions
- Operation by button for system evaluation with precautions
- One-For-All Tool for multiple CAN based DTI Devices
- Online firmware update capability.

The functionality is expanding specifically for the F-SIC kit with a thermal and heat simulation. This feature aids Formula Student teams in determining the thermal dissipation of the inverter and the motor, including transient simulation. It also allows users to see the required coolant temperature for different load cycles.

To use this feature, follow these steps:

- Download the latest DTI CAN Tool graphical interface from our website.
- After installation, you do not need to connect the DTI diagnostic tool to the COM port; simply close the pop-up window and select the "Simulation" function from the left menu bar.
- Further instructions will then be visible

> It is important that the uploaded CSV file matches the format, number of columns, and units as per the provided template. The CSV file can contain negative current

values, but they must not exceed the inverter's maximum rating. If any value exceeds the maximum rating, it will be capped.

If the inverter temperature exceeds the limit value, the inverter will begin to limit, and any AC current values above that threshold will no longer be achievable. If the AC current decreases again, operations will resume without limitations.

9. WIRING THE CONTROLLER

9.1 Overview

Digital signal input and output are available for switches, sensors, contactors, and hydraulic valves, and CAN communication for controlling and configure the device.

The wiring must be carried out by a qualified person.

Special attention must be paid to the equipotential bonding for components which are connected to the unit and which do not have isolated inputs and outputs (equalizing connection GND). The equalizing currents may damage components and parts permanently.

> The units, the inductive and capacitive accessories as well as the power wiring can generate strong electric and electromagnetic fields. These fields may be dangerous for persons having electronic medical aids or appliances (e.g.: cardiac pacemakers). Sufficient distance to these electrical parts must be observed. The switch cabinet must be labelled accordingly.

Keep all wiring harnesses short and route wiring close to vehicle metalwork. Keep all signal wires clear of power cables and consider the use of screened cable. Keep control wiring clear of power cables when it carries analogue information - for example, accelerator wiring. Tie all wiring securely.

> **Do not apply power until you are certain the controller high power and signal wiring is correct and has been double checked. Wiring faults will damage the controller.**

9.2 Harness connector wiring

You must use MOLEX connector parts. The harness side connector type: **5054321201** and the corresponding crimp pins: **505431-1100** for 28 AWG wire.

A special hand crimp tool is required for crimping the pins: **2002180300**

Try to create the simplest and shortest wiring harness as possible. You can involve the harness with cable sleeving polyamide or polypropylene spiral, you can protect it from external influences.

9.2.1 CAN Wiring

CAN1 is required for firmware updates using the **DTI-COM diagnostic tool**. The tool is not based on OBD-II standard, only the connector.

CAN 1 is used for firmware updates and configuration; CAN 2 is used for controlling the inverter.

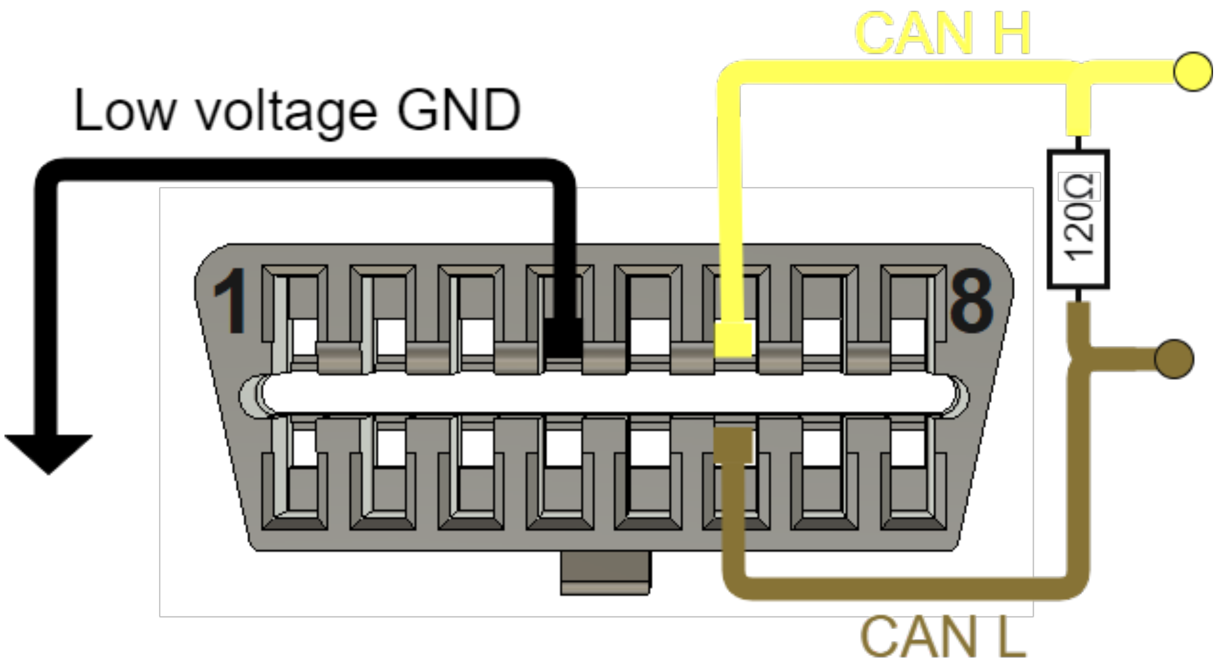


Figure 11. OBD connector pinout for CAN Bus

Apply 120Ω resistors are required for both ends of the CAN bus.

9.2.2 Switchable Internal CAN termination resistor

We installed switchable CAN termination resistors in the inverter to reduce the complexity of the cable harness. By default, both switches are turned *OFF* (Not terminated bus)

> It is important to de-energize the system before disassembling and activating the switches. It is essential that termination resistors are only present at the two ends of the CAN bus. If the four inverters are on a common CAN bus, only the termination resistor of the last inverter should be switched on, and the termination resistor on the other side of the bus should be installed.

To switch it, the top cover plate, which is secured by four screws, must be removed. Under this cover, there are two DIP switches.

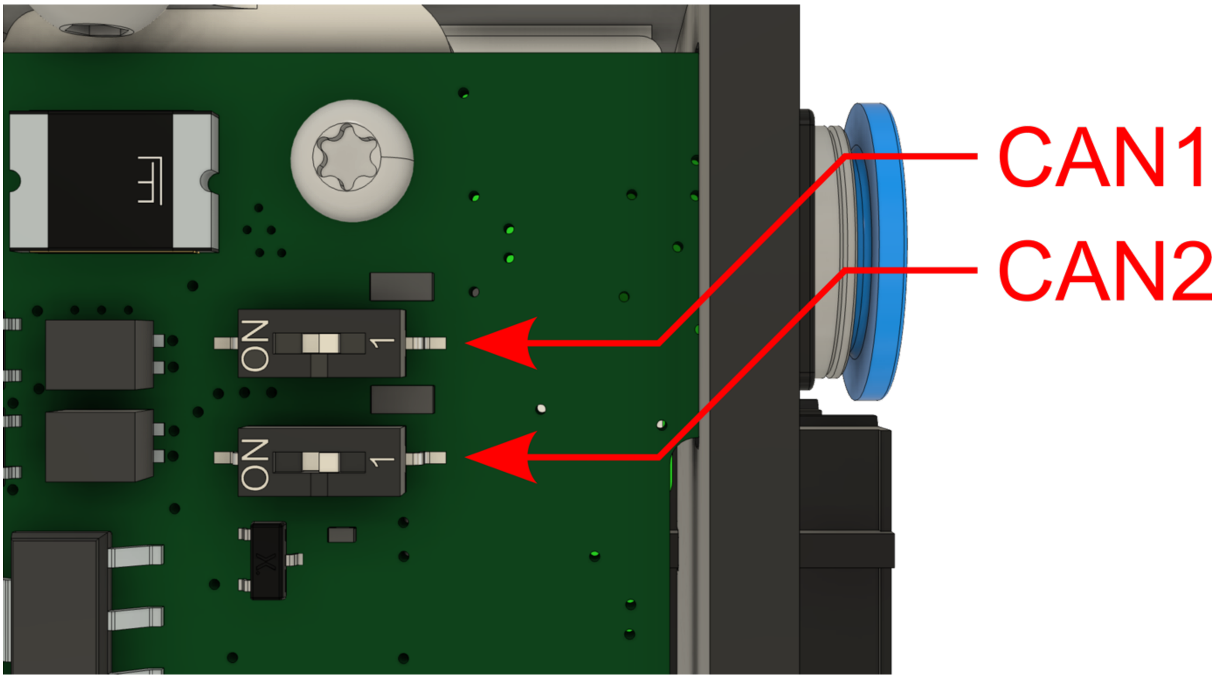


Figure 12. CAN Bus termination resistor switches

> Pay close attention to maintaining a clean environment when removing the cover. Use ESD-protected tools and work in an ESD-safe area. Ensure no contaminants enter the inverter, as this could cause malfunction.

10. ELECTRICAL CONNECTION SCHEMATICS

10.1 Input supply

- Reverse polarity protection
- Overvoltage protection
- Resettable overcurrent protection

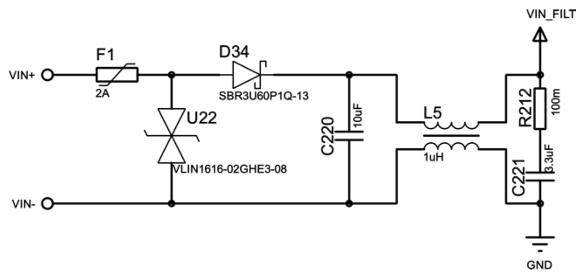


Figure 13. Input supply schematic diagram

10.2 Digital input

- Button debounce
- Overvoltage protection
- Internal pullup resistor

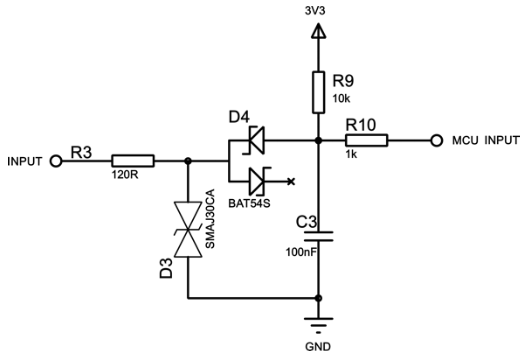


Figure 14. Digital input schematic diagram

10.3 Digital output

- Maximum 24V
- High-Speed resettable fuse
- Maximum 240 mA holding current
- Internal freewheeling diode

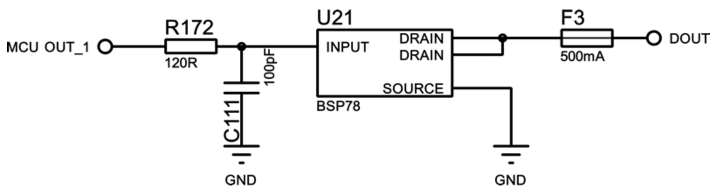


Figure 15. Digital output schematic diagram

10.4 CAN peripheral

- Common mode choke filter
- ESD protection
- Switchable termination resistor

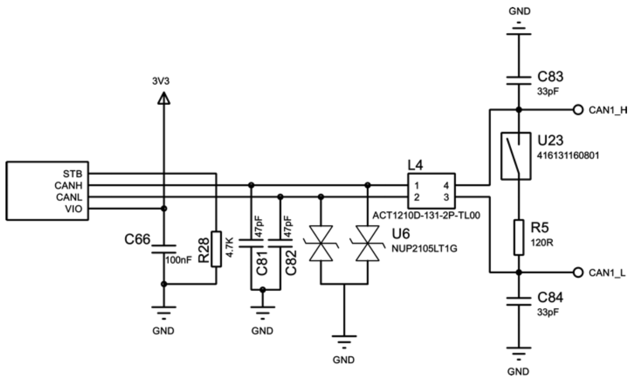


Figure 16. CAN peripheral schematic diagram

10.5 EnDat peripheral

- Common mode choke filter
- ESD protection

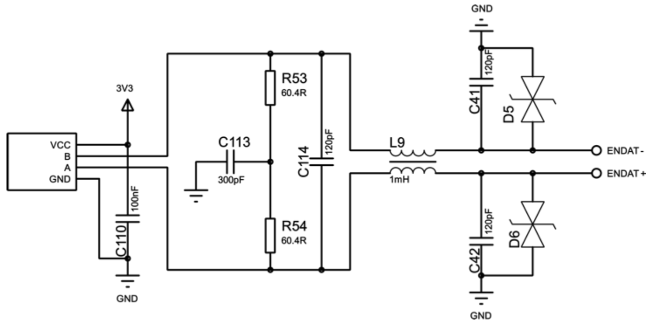


Figure 17. EnDat peripheral schematic diagram

10.6 Encoder peripheral

- Internal termination resistor
- ESD protection

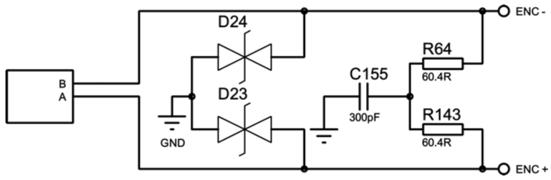


Figure 18. Encoder peripheral schematic diagram

11. LIQUID COOLING

To minimize pressure drop, it is recommended to connect the liquid cooling of the two inverter blocks in parallel. It is important to ensure equal water flow through both inverters, so follow the structure shown in the image when connecting them in parallel.

The F-SiC-A inverter is fitted with 2 pieces of **NPQH-D-G14-Q10-P10**(∅10) cooling outlet.

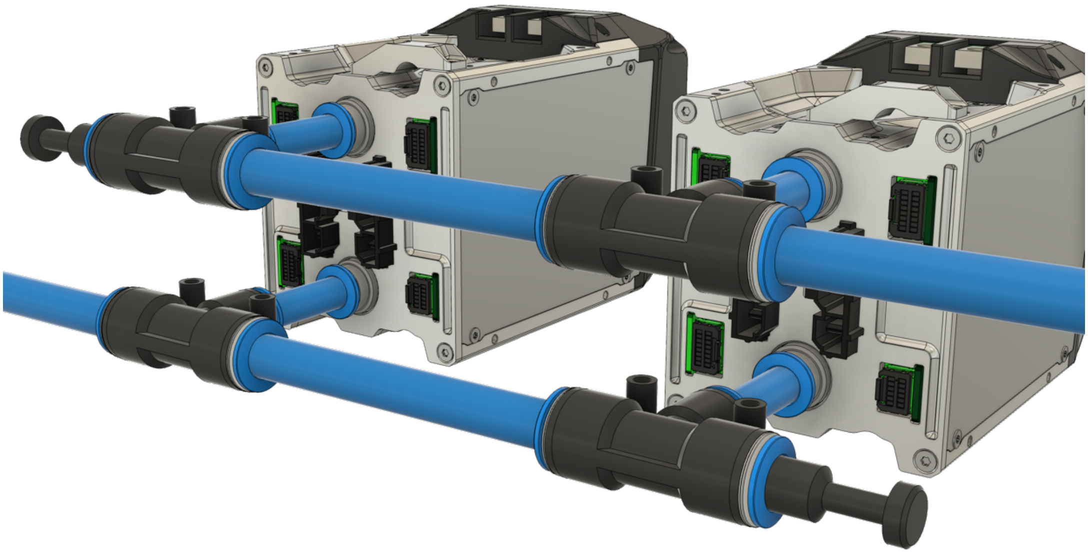


Figure 19. 4WD cooling system layout

In the schematic shown in the image, a 12 mm tube is used in the main cooling line.

T connector (reducing from ∅12 to ∅10): **NPQH-T-Q12-Q10-P10.**

The unused ends are sealed with Festo **NPQH-P-S12-P10** (∅12) blanking plugs.

Recommended fittings: Festo NPQH series, up to 150 °C and a pressure range up to 20 bar.

Recommended tubing: **Festo PTFEN-12X2-NT** (∅12) and **Festo PTFEN-10X1,5-NT** (∅10)

12. LIST OF TABLES

13. LIST OF FIGURES